TIC 405529940

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        1.026596 d
  Epoch (t0)    2825.6009 BTJD
  Duration      0.0840 d
  Depth         2.998 %  (29,976 ppm)
  SNR           85.4

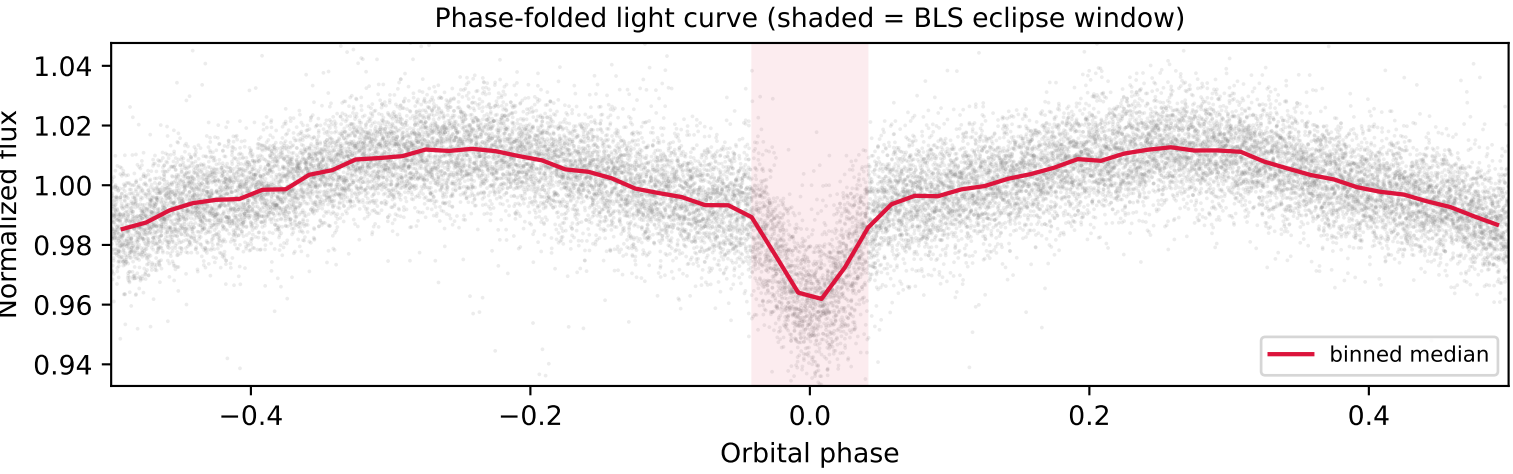
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      52  (gate >= 3)
  Robust depth ratio     1.03  (gate >= 0.50)
  Robust depth sigma     80.9  (gate >= 7.0)
  Odd/even z-score       2.42  (consistent)
  Secondary eclipse      not detected
```

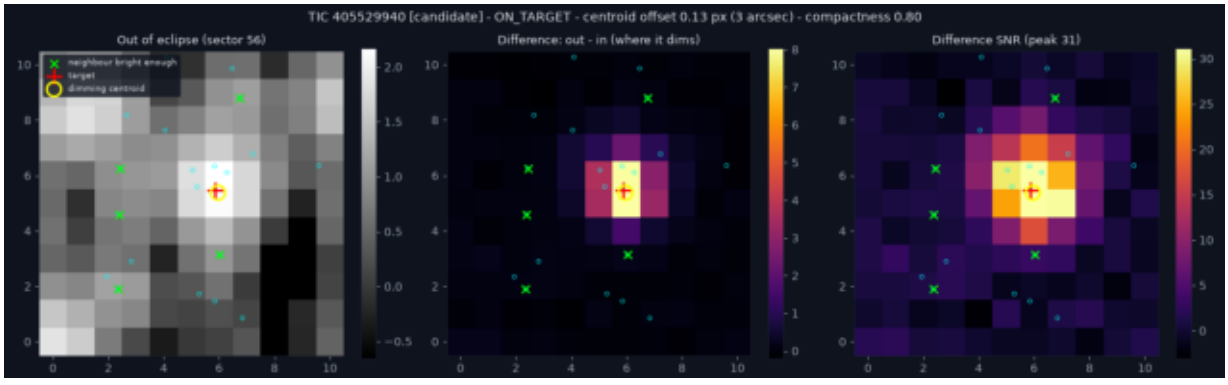
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 56    light-curve dip in this sector: 85.0 sigma    t0 refined by 0.0100 phase
  Difference-image peak SNR   31.16   (gate >= 5.0)      Compactness   0.80   (gate >= 0.50)
  Dimming centroid offset     0.13 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 118154236
    2.24 px away, dTmag +3.21 - farther from the dimming than the target
```





No caveats - clean detection on every check.